

Global Retail

Sales Analysis

Agenda

- 01.** Data Understanding
- 02.** Data_Cleaning
- 03.** Data_Modeling
- 04.** Product_Analysis
- 05.** Customer Analysis
- 06.** Store & Sales Analysis
- 07.** Financial Analysis
- 08.** Executive Dashboard & KPI Analysis



Data Understanding

SELECT * FROM Customers;

SELECT * FROM Products;

SELECT * FROM Sales;

SELECT * FROM Stores;

SELECT * FROM Exchange_Rates;

SELECT * FROM Data_Dictionary;

Results										
	CustomerKey	Gender	Name	City	State_Code	State	Zip_Code	Country	Continent	Birthday
1	301	Female	Lilly Harding	WANDEARAH EAST	SA	South Australia	5523	Australia	Australia	1939-03-07
2	325	Female	Madison Hull	MOUNT BUDD	WA	Western Australia	6522	Australia	Australia	1979-09-27
3	554	Female	Claire Ferres	WINJALLOK	VIC	Victoria	3380	Australia	Australia	1947-05-26
4	786	Male	Jai Poltpalingada	MIDDLE RIVER	SA	South Australia	5223	Australia	Australia	1957-09-17
5	1042	Male	Aidan Pankhurst	TAWONGA SOUTH	VIC	Victoria	3698	Australia	Australia	1965-11-19

	ProductKey	Product_Name	Brand	Color	Unit_Cost_USD	Unit_Price_USD	SubcategoryKey	Subcategory	CategoryKey	Category
1	1	Contoso 512MB MP3 Player E51 Silver	Contoso	Silver	6.62	12.99	101	MP4&MP3	1	Audio
2	2	Contoso 512MB MP3 Player E51 Blue	Contoso	Blue	6.62	12.99	101	MP4&MP3	1	Audio
3	3	Contoso 1G MP3 Player E100 White	Contoso	White	7.40	14.52	101	MP4&MP3	1	Audio
4	4	Contoso 2G MP3 Player E200 Silver	Contoso	Silver	11.00	21.57	101	MP4&MP3	1	Audio

	Order_Number	Line_Item	Order_Date	Delivery_Date	CustomerKey	StoreKey	ProductKey	Quantity	Currency_Code
1	366000	1	2016-01-01	NULL	265598	10	1304	1	CAD
2	366001	1	2016-01-01	2016-01-13	1269051	0	1048	2	USD
3	366001	2	2016-01-01	2016-01-13	1269051	0	2007	1	USD
4	366002	1	2016-01-01	2016-12-01	266019	0	1106	7	CAD
5	366002	2	2016-01-01	2016-12-01	266019	0	373	1	CAD

	StoreKey	Country	State	Square_Meters	Open_Date
3	3	Australia	South Australia	2000	2012-01-07
4	4	Australia	Tasmania	2000	2010-01-01
5	5	Australia	Victoria	2000	2015-12-09
6	6	Australia	Western Australia	2000	2010-01-01

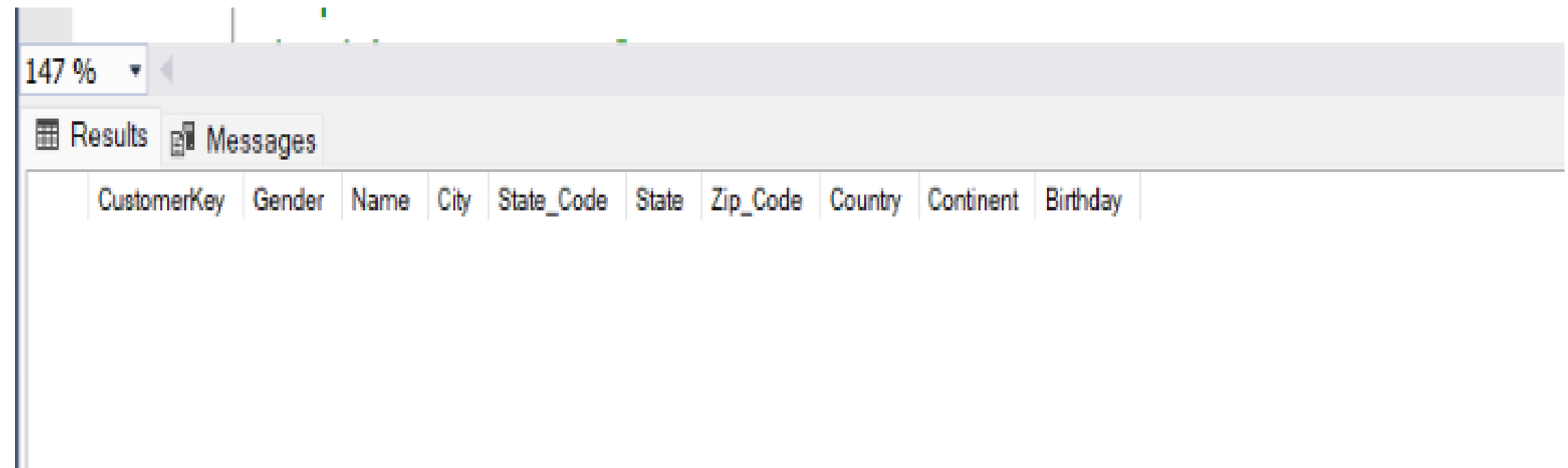
	Date	Currency	Exchange	Sql_Date_Format
3	2015-01-01	AUD	1.22140002250...	2015-01-01
4	2015-01-01	EUR	0.82370001077...	2015-01-01
5	2015-01-01	GBP	0.64149999618...	2015-01-01
6	2015-02-01	USD	1	2015-02-01

	Table	Field	Description
4	Sales	Delivery Date	Date the order was delivered
5	Sales	CustomerKey	Unique key identifying which customer placed t...
6	Sales	StoreKey	Unique key identifying which store processed th...
7	Sales	ProductKey	Unique key identifying which product was purch...
8	Sales	Quantity	Number of items purchased
9	Sales	Currency Code	Currency used to process the order

DATA VALIDATION

Checking NULL Values in Customers Table

```
SELECT *  
FROM Customers  
WHERE CustomerKey IS NULL  
OR Name IS NULL  
OR Gender IS NULL  
OR City IS NULL  
OR State_Code IS NULL  
OR State IS NULL  
OR Zip_Code IS NULL  
OR Country IS NULL  
OR Continent IS NULL  
OR Birthday IS NULL;
```



CustomerKey	Gender	Name	City	State_Code	State	Zip_Code	Country	Continent	Birthday
-------------	--------	------	------	------------	-------	----------	---------	-----------	----------

Data Cleaning

- Data Type Validation
- ZIP Code Validation
- NULL Handling
- Missing Value Verification
- Duplicate Validation
- Exchange Rate Validation
- Date Format Validation
- No major data cleaning was required because the dataset was already structured.

Data Modeling

Fact Table – Sales

Reason for selecting Sales as the Fact Table:

- Stores every sales transaction (one record per order).
- Contains measurable business metrics such as Quantity, Revenue, and Order Details.
- Connects all business entities using CustomerKey, ProductKey, and StoreKey.
- Acts as the central table for business analyses.
- Used to calculate KPIs such as Revenue, Profit, Orders, and Sales Trends.

Dimension Tables

Customers

- Provides customer information for customer behavior and segmentation analysis.

Products

- Provides product attributes (Category, Brand, Cost, Price) for product performance and profitability analysis.

Stores

- Provides store location details for store performance and delivery analysis.

Exchange_Rates

- Provides currency conversion rates to calculate revenue in local currency for global sales analysis.

Customer Analysis

Write a query to display every sales order along with the corresponding customer name, product name, and quantity purchased.

```
SELECT
    Sales.[Order_Number],
    Customers.[Name],
    Products.[Product_Name],
    Sales.Quantity
FROM Sales
INNER JOIN Customers
    ON Sales.CustomerKey = Customers.CustomerKey
INNER JOIN Products
    ON Sales.ProductKey = Products.ProductKey;
```

	Order_Number	Name	Product_Name	Quantity
1	366000	Tyler Vaught	Contoso Lens Adapter M450 White	1
2	366001	Frank Upchurch	A. Datum SLR Camera X136 Silver	2
3	366001	Frank Upchurch	Fabrikam Microwave 1.5CuFt X1100 Black	1
4	366002	Joan Roche	Contoso SLR Camera M146 Orange	7
5	366002	Joan Roche	Adventure Works Laptop8.9 E0890 White	1
6	366002	Joan Roche	Contoso SLR Camera X143 Silver	4
7	366004	Sam Nelson	Adventure Works 52" LCD HDTV X790W White	6
8	366004	Sam Nelson	The Phone Company PDA Handheld 3.5 inch M610 Black	2
9	366005	Ouassim Heuver	Adventure Works Desktop PC3.0 MS300 Silver	4
10	366007	Damiana Corona	Contoso DVD Player M110 Silver	1
11	366007	Damiana Corona	Proseware Office Jet All-in-One Printer M500 Black	5
12	366008	Callisto Lo Duca	Fabrikam Trendsetter 1/3" 8.5mm X200 Blue	9
13	366008	Callisto Lo Duca	Contoso Original K1m Li-Ion Standard Battery E170 White	1
14	366008	Callisto Lo Duca	A. Datum SLR Camera X139 Gold	5
15	366009	Hazel Skiles	Contoso DVD 7-Inch Player Portable E200 Black	1
16	366010	Carlota Etzel	WWI Screen 113in M1610 Black	5
17	366011	Mike Mogensen	Adventure Works 19" Color Digital TV E35 Brown	7
18	366011	Mike Mogensen	Contoso DVD 58 DVD Storage Binder M55 Red	1
19	366011	Mike Mogensen	Contoso DVD 38 DVD Storage Binder E25 Silver	1
20	366011	Mike Mogensen	Proseware Air conditioner 10000BTU M490 Red	3
21	366012	Rebecca Koch	MGS Dungeon Siege 2009 E163	1
22	366012	Rebecca Koch	SV DVD 38 DVD Storage Binder E25 Red	4
23	366012	Rebecca Koch	Contoso USB Cable M250 White	1
24	366013	Max Norris	Contoso DVD 14-Inch Player Portable L100 Silver	5
25	366014	Clara Toscani	Adventure Works 19" Portable LCD HDTV M110 Black	1
26	366014	Clara Toscani	SV Hand Games women M40 Red	2

Customer Analysis

Write a query to identify customers who have purchased products from more than one product category.

```
SELECT
    Customers.Name,
    COUNT(DISTINCT Products.Category) AS
        CategoriesPurchased
FROM Sales
INNER JOIN Customers
    ON Sales.CustomerKey = Customers.CustomerKey
INNER JOIN Products
    ON Sales.ProductKey = Products.ProductKey
GROUP BY
    Customers.Name
HAVING
    COUNT(DISTINCT Products.Category) > 1
```

	Name	CategoriesPurchased
1	Aagje Maring	6
2	Aaron Booth	5
3	Aaron Brock	3
4	Abbey Finch	4
5	Abbie Haynes	4
6	Abby Plowman	7
7	Abigail Anderson	4
8	Abigail Armstrong	4
9	Aceline Moise	5
10	Ada Hanlon	2
11	Adel Gustafsson	2
12	Adolfo Genovesi	7
13	Adrienne Soriano	4
14	Adrian Mullen	4

Product Analysis

Write a query to find the best-selling product (by revenue) within each product category.

```
WITH ProductRevenue AS
(
    SELECT
        Products.Category,
        Products.[Product_Name],
        SUM(Sales.Quantity * Products.[Unit_Price_USD])
        AS Sales,
        ROW_NUMBER() OVER
        (PARTITION BY Products.Category
        ORDER BY SUM(Sales.Quantity * Products.[Unit_Price_USD])
        DESC) AS RN
    FROM Sales
    INNER JOIN Products
        ON Sales.ProductKey = Products.ProductKey
    GROUP BY
        Products.Category,
        Products.[Product_Name]
)
SELECT Category, [Product_Name], Sales, RN
FROM ProductRevenue
WHERE RN = 1
```

	Category	Product_Name	Sales	RN
1	Audio	WWI 4GB Video Recording Pen X200 Yellow	109520.00	1
2	Cameras and camcorders	Fabrikam Independent Filmmaker 1/3" 8.5mm X200 Grey	132600.00	1
3	Cell phones	The Phone Company Touch Screen Phone 1600 TFT-1....	118389.00	1
4	Computers	WWI Desktop PC2.33 X2330 Black	505450.00	1
5	Games and Toys	MGS Age of Empires Expansion: The Rise of Rome X900	48502.80	1
6	Home Appliances	Contoso Water Heater 7.2GPM X1800 Silver	231575.00	1
7	Music, Movies and Audio Books	SV DVD 14-Inch Player Portable L100 Silver	101656.09	1
8	TV and Video	Adventure Works 52" LCD HDTV X590 White	394398.64	1

Product Analysis

Finance wants to know which product categories generate the highest profit margin. Write a query to display category, revenue, cost, profit, and profit percentage, sorted by profit percentage in descending order.

```
SELECT Products.Category,
SUM(Sales.Quantity * Products.[Unit_Price_USD]) AS Revenue,
SUM(Sales.Quantity * Products.[Unit_Cost_USD]) AS Cost,
SUM((Sales.Quantity * Products.[Unit_Cost_USD]) -
(Sales.Quantity * Products.[Unit_Cost_USD])) AS Profit,
(
SUM(
(Sales.Quantity * Products.[Unit_Cost_USD])
-
(Sales.Quantity * Products.[Unit_Cost_USD])
)
*100.0
/
NULLIF
(
SUM(
Sales.Quantity * Products.[Unit_Price_USD]
),
0
)
) AS ProfitPercentage
FROM Sales
INNER JOIN Products
ON Sales.ProductKey = Products.ProductKey
GROUP BY Products.Category
ORDER BY ProfitPercentage DESC;
```

	Category	Revenue	Cost	Profit	ProfitPercentage
1	TV and Video	5928982.69	2392288.30	0.00	0.0000000000000000
2	Audio	3169627.74	1341775.97	0.00	0.0000000000000000
3	Home Appliances	10795478.59	4070810.86	0.00	0.0000000000000000
4	Computers	19301595.46	8024147.56	0.00	0.0000000000000000
5	Cell phones	6183791.22	2685164.68	0.00	0.0000000000000000
6	Games and Toys	724829.43	328160.66	0.00	0.0000000000000000
7	Cameras and camcorders	6520168.02	2600367.03	0.00	0.0000000000000000
8	Music, Movies and Audio Books	3131006.44	1221747.27	0.00	0.0000000000000000

Store Analysis

The Operations Manager wants to identify which stores consistently deliver orders the fastest. Write a query to display store key, country, state, total orders, and average delivery days, sorted by fastest delivery time.

```
SELECT
    Stores.StoreKey, Stores.Country, Stores.State,
    COUNT(Sales.[Order_Number]) AS TotalOrders,
    AVG(
        DATEDIFF
        (
            DAY,
            Sales.[Order_Date],
            Sales.[Delivery_Date]
        )
    ) AS AvgDeliveryDays
FROM Sales
INNER JOIN Stores
ON Sales.StoreKey = Stores.StoreKey
GROUP BY Stores.StoreKey, Stores.Country, Stores.State
ORDER BY
AvgDeliveryDays asc
```

	StoreKey	Country	State	TotalOrders	AvgDeliveryDays
1	1	Australia	Australian Capital Territory	292	NULL
2	2	Australia	Northern Territory	18	NULL
3	4	Australia	Tasmania	431	NULL
4	5	Australia	Victoria	892	NULL
5	6	Australia	Western Australia	615	NULL
6	8	Canada	Newfoundland and Labrador	1360	NULL
7	9	Canada	Northwest Territories	1577	NULL
8	10	Canada	Nunavut	1269	NULL
9	12	France	Basse-Normandie	224	NULL
10	13	France	Corse	184	NULL
11	14	France	Franche-Comté	126	NULL
12	15	France	La Réunion	235	NULL
13	16	France	Limousin	214	NULL
14	17	France	Martinique	210	NULL
15	18	France	Mayotte	194	NULL
16	19	Germany	Berlin	635	NULL
17	20	Germany	Brandenburg	443	NULL
18	21	Germany	Freie Hansestadt Bremen	391	NULL
19	22	Germany	Freistaat Thüringen	800	NULL
20	23	Germany	Hamburg	674	NULL
21	24	Germany	Hessen	703	NULL
22	26	Germany	Saarland	295	NULL
23	27	Germany	Sachsen-Anhalt	721	NULL
24	28	Italy	Caltanissetta	214	NULL
25	29	Italy	Enna	1011	NULL

Store Analysis

Management wants to investigate stores with poor delivery performance. Write a query to display the top 5 stores with the highest average delivery days.

```
SELECT TOP (5)
    Stores.StoreKey, Stores.Country, Stores.State,
    AVG
    (
        DATEDIFF
        (
            DAY,
            Sales.[Order_Date],
            Sales.[Delivery_Date]
        )
    ) AS AvgDeliveryDays
FROM Sales
INNER JOIN Stores
ON Sales.StoreKey = Stores.StoreKey
GROUP BY Stores.StoreKey, Stores.Country, Stores.State
ORDER BY AvgDeliveryDays DESC;
```

	StoreKey	Country	State	AvgDeliveryDays
1	0	Online	Online	49
2	6	Australia	Western Australia	NULL
3	5	Australia	Victoria	NULL
4	4	Australia	Tasmania	NULL
5	2	Australia	Northern Territory	NULL

Sales Analysis

The CEO wants to track cumulative revenue throughout the year. Write a query to display year, month, monthly revenue, and running (cumulative) revenue.

```
WITH MonthlyRevenue AS
(
    SELECT
        YEAR(Sales.[Order_Date]) AS SalesYear,
        MONTH(Sales.[Order_Date]) AS SalesMonth,
        SUM
        (
            Sales.Quantity * Products.[Unit_Price_USD]
        ) AS Revenue
    FROM Sales
    INNER JOIN Products
        ON Sales.ProductKey = Products.ProductKey
    GROUP BY YEAR(Sales.[Order_Date]), MONTH(Sales.[Order_Date])
)
SELECT SalesYear, SalesMonth, Revenue,
    SUM(Revenue) OVER (ORDER BY SalesYear, SalesMonth) AS RunningRevenue
FROM MonthlyRevenue
ORDER BY SalesYear, SalesMonth;
```

	SalesYear	SalesMonth	Revenue	RunningRevenue
1	NULL	NULL	33910829.80	33910829.80
2	2016	1	290267.44	34201097.24
3	2016	2	302376.86	34503474.10
4	2016	3	311468.16	34814942.26
5	2016	5	218976.22	35033918.48
6	2016	6	202782.24	35236700.72
7	2016	7	189115.38	35425816.10
8	2016	8	215973.22	35641789.32
9	2016	9	238066.73	35879856.05
10	2016	10	256887.22	36136743.27
11	2016	11	276704.51	36413447.78
12	2016	12	231636.89	36645084.67
13	2017	1	364307.77	37009392.44
14	2017	2	218467.93	37227860.37
15	2017	3	269570.56	37497430.93
16	2017	5	237336.33	37734767.26
17	2017	6	273449.64	38008216.90
18	2017	7	218126.17	38226343.07
19	2017	8	197963.94	38424307.01
20	2017	9	230014.95	38654321.96
21	2017	10	249466.01	38903787.97
22	2017	11	235505.66	39139293.63
23	2017	12	274254.77	39413548.40
24	2018	1	510666.51	39924214.91
25	2018	2	360016.11	40284231.02

Sales Analysis

Management wants to compare each month's revenue with the previous month. Write a query to display year, month, current month revenue, previous month revenue, and month-over-month growth.

```
WITH MonthlyRevenue AS
(
    SELECT YEAR(Sales.[Order_Date]) AS SalesYear,
           MONTH(Sales.[Order_Date]) AS SalesMonth,
           SUM
             (Sales.Quantity * Products.[Unit_Price_USD]) AS Revenue
    FROM Sales
    INNER JOIN Products
        ON Sales.ProductKey = Products.ProductKey
    GROUP BY YEAR(Sales.[Order_Date]),
             MONTH(Sales.[Order_Date])
)
SELECT SalesYear, SalesMonth, Revenue,
       LAG(Revenue)
       OVER(ORDER BY SalesYear, SalesMonth) AS PreviousMonthRevenue,
       Revenue
       -
       LAG(Revenue)
       OVER ( ORDER BY SalesYear, SalesMonth ) AS MoMGrowth
FROM MonthlyRevenue
ORDER BY SalesYear, SalesMonth
```

	SalesYear	SalesMonth	Revenue	PreviousMonthRevenue	MoMGrowth
1	NULL	NULL	33910829.80	NULL	NULL
2	2016	1	290267.44	33910829.80	-33620562.36
3	2016	2	302376.86	290267.44	12109.42
4	2016	3	311468.16	302376.86	9091.30
5	2016	5	218976.22	311468.16	-92491.94
6	2016	6	202782.24	218976.22	-16193.98
7	2016	7	189115.38	202782.24	-13666.86
8	2016	8	215973.22	189115.38	26857.84
9	2016	9	238066.73	215973.22	22093.51
10	2016	10	256887.22	238066.73	18820.49
11	2016	11	276704.51	256887.22	19817.29
12	2016	12	231636.89	276704.51	-45067.62
13	2017	1	364307.77	231636.89	132670.88
14	2017	2	218467.93	364307.77	-145839.84
15	2017	3	269570.56	218467.93	51102.63
16	2017	5	237336.33	269570.56	-32234.23
17	2017	6	273449.64	237336.33	36113.31
18	2017	7	218126.17	273449.64	-55323.47
19	2017	8	197963.94	218126.17	-20162.23
20	2017	9	230014.95	197963.94	32051.01
21	2017	10	249466.01	230014.95	19451.06
22	2017	11	235505.66	249466.01	-13960.35
23	2017	12	274254.77	235505.66	38749.11
24	2018	1	510666.51	274254.77	236411.74
25	2018	2	360016.11	510666.51	-150650.40
26	2018	3	333754.50	360016.11	-26261.61
27	2018	5	453991.28	333754.50	120236.78

Profit Analysis

Write a query to calculate revenue, cost, profit, profit percentage, and revenue in local currency for each country.

```
SELECT
    Customers.Country,
    SUM(Sales.Quantity * Products.[Unit_Price_USD]) AS RevenueUSD,
    SUM(Sales.Quantity * Products.[Unit_cost_USD]) AS CostUSD,
    SUM((Sales.Quantity * Products.[Unit_Price_USD])
        -(Sales.Quantity * Products.[Unit_cost_USD]))
    ) AS ProfitUSD,
    (
        SUM(
            (Sales.Quantity * Products.[Unit_Price_USD])
            -
            (Sales.Quantity * Products.[Unit_cost_USD])
        )
        /
        NULLIF(
            SUM(Sales.Quantity * Products.[Unit_Price_USD]),
            0
        )
    ) * 100 AS ProfitPercentage,
    SUM(
        Sales.Quantity
        * Products.[Unit_Price_USD]
        * Exchange_Rates.Exchange
    ) AS RevenueLocalCurrency
FROM Sales
INNER JOIN Customers
    ON Sales.CustomerKey = Customers.CustomerKey
INNER JOIN Products
    ON Sales.ProductKey = Products.ProductKey
INNER JOIN Exchange_Rates
    ON Sales.[Currency_Code] = Exchange_Rates.Currency
    AND Sales.[Order_Date] = Exchange_Rates.Date
GROUP BY
    Customers.Country
```

	Country	RevenueUSD	CostUSD	ProfitUSD	ProfitPercentage	RevenueLocalCurrency
1	Italy	1012481.96	412901.83	580380.19	57.32	892625.213771585
2	Netherlands	758724.00	299862.72	430061.37	56.68	666706.649757853
3	Germany	2039925.71	817192.11	1174733.75	57.58	1795804.91616994
4	United States	11436958.40	4659715.22	6559643.86	57.35	11436958.4000006
5	Australia	1101120.16	425889.95	617630.39	56.09	1525687.34513581
6	United Kingdom	2612192.42	1081275.58	1476517.01	56.52	2002966.73058886
7	Canada	1950401.88	790486.99	1092715.10	56.02	2567745.12426814
8	France	556237.62	229533.29	326704.33	58.73	490556.760857946

Executive Analysis

Marketing wants to know which brands dominate each country. Write a query to display the top 5 brands (by revenue) within each country, along with their rank.

```
WITH BrandRevenue AS
(
    SELECT Customers.Country, Products.Brand,
           SUM(
               Sales.Quantity * Products.[Unit_Price_USD]
           ) AS Revenue,
           ROW_NUMBER() OVER ( PARTITION BY Customers.Country ORDER BY
                               SUM(Sales.Quantity * Products.[Unit_Price_USD])
                               DESC) AS RN
    FROM Sales
    INNER JOIN Customers
    ON Sales.CustomerKey = Customers.CustomerKey
    INNER JOIN Products
    ON Sales.ProductKey = Products.ProductKey
    GROUP BY
        Customers.Country, Products.Brand
)
SELECT COUNTRY, BRAND, REVENUE, RN
FROM BrandRevenue
WHERE RN <=5
ORDER BY Country, RN;
```

	COUNTRY	BRAND	REVENUE	RN
1	Australia	Adventure Works	587074.33	1
2	Australia	Contoso	545630.94	2
3	Australia	Wide World Importers	398695.79	3
4	Australia	Fabrikam	354337.92	4
5	Australia	The Phone Company	238015.00	5
6	Canada	Adventure Works	957676.08	1
7	Canada	Contoso	944794.02	2
8	Canada	Wide World Importers	730311.23	3
9	Canada	Fabrikam	579131.76	4
10	Canada	The Phone Company	416541.00	5
11	France	Adventure Works	308806.70	1
12	France	Contoso	308594.43	2
13	France	Wide World Importers	258811.12	3
14	France	Fabrikam	192738.81	4
15	France	The Phone Company	145336.00	5
16	Germany	Adventure Works	1187321.10	1
17	Germany	Contoso	1060641.09	2
18	Germany	Wide World Importers	984727.10	3

Sales Performance Overview

55755479.59

Total Revenue

26326

Total Orders

2117.8864

Average Order
Value

31798221.30

Total Profit

Key Findings

01.



Revenue increased compared to the previous period.

02.



Several regions demonstrated significant market potential.

03.



Digital sales channels delivered the highest performance.

04.



Top-selling products continued to drive growth.

Action Items

01.



Strengthen customer retention initiatives.

02.



Expand marketing activities in high-growth regions.

03.



Optimize product inventory planning.

04.



Improve performance monitoring across sales channels.

Presented by:
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Global Retail

Thank You

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